Conjecture 00000001385: an infinite Hilbert spectral-count plateau

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1 A necessary jump clause fails

The source defines $N_T(\lambda)$ as a spectral counting *function* and requires that after an infinite value, all higher values equal $2^{\aleph_0}$. With the standard Hilbert dimension of spectral subspaces, this necessary clause is false. Take the genuine probability circle

$$X = \mathbb{R}/\mathbb{Z}, \qquad \mu = \text{normalized Haar measure}, \qquad T(x) = x.$$

Its actual Koopman operator on $E = L^2(X, \mu; \mathbb{C})$ is $Uf = f \circ T = f$. Its only eigenvalue is the real number 1, and the eigenspace is all of E. The genuine Fourier Hilbert basis is indexed by $\mathbb{Z}$. Consequently

$$N_T(1) = \aleph_0 = N_T(2) \neq 2^{\aleph_0}.$$

Here 1 and 2 are real *thresholds*; 2 is not asserted to be a second spectral point. Thus the infinite-value jump clause, and hence the source's conjunction under this standard function reading, is disproved. We do not need to determine its other claims about alternation or spectral-measure atomicity.

2 Actual transformations and cutoff subspaces

Lean constructs Mathlib's actual `AddCircle 1`, its Haar probability measure, and the actual `Lp` space with exponent 2. The map T is proved measure preserving and involutive. The operator U is the actual `Lp.compMeasurePreserving` construction and its complex-linear refinement, not an assigned diagonal model. The proof establishes $U = \mathrm{id}$, existence of eigenvalue 1, and the absence of any other eigenvalue.

General complex eigenvalues have no specified order in the source. This example avoids inventing one: its spectrum consists only of the real eigenvalue 1. The formal real-eigenvalue cutoff is the genuine closed subspace

$$S_a = \overline{\sum_{r \in \mathbb{R},\ r \leq a} \ker(U - rI)}.$$

For robustness, the package independently constructs the modulus cutoff

$$M_a = \overline{\sum_{z \in \mathbb{C},\ |z| \leq a} \ker(U - zI)}.$$

For every real $a \geq 1$, the actual eigenspace at 1 is one of the summands and already equals E, so $S_a = M_a = E$. Both cutoff interpretations give the displayed counterexample.

3 Dimension is proved from genuine Hilbert bases

Mathlib's actual Fourier Hilbert basis is used with all its orthonormality and completeness proofs. The formal argument also proves that *every* Hilbert basis of E has index cardinality $\aleph_0$, so its dimension is independent of the chosen Fourier enumeration. If $(b_i)_{i\in I}$ is another Hilbert basis, then for each Fourier vector e_n the support of its square-summable coordinate vector in (b_i) is countable. The union of those supports over $n \in \mathbb{Z}$ is all of I: otherwise a nonzero b_i would be orthogonal to every Fourier vector, contradicting Fourier completeness. Thus I is countable. It cannot be finite, because that would make E finite dimensional despite the infinite linearly independent Fourier family. Hence $|I| = \aleph_0$.

The package defines the Hilbert dimension of a subspace by the infimum of the cardinalities admitting an actual Hilbert basis. For the relevant full subspace it proves that this set is the nonempty singleton $\{\aleph_0\}$, using an explicitly transported Fourier basis for existence and the preceding uniqueness proof. There is no empty-infimum fallback, no assigned value for N_T, and no substitution of Hamel `Module.rank` for Hilbert dimension. The two counting functions are obtained by applying this dimension definition to the actual cutoff subspaces. Cantor's strict inequality $\aleph_0 < 2^{\aleph_0}$ closes both formal negations of the required jump property.

4 Conventions, primary references and verification

The Hilbert interpretation is standard in this setting. S. New's Waterloo *Chapter 2: Hilbert Spaces*, PDF/printed page 10, Theorem 2.35 and Definition 2.36, proves basis-cardinality uniqueness and explicitly uses Hilbert dimension for a Hilbert space unless otherwise specified, distinguished from Hamel dimension. B. Solomyak's *Ergodic Theory* notes, 21 November 2019, PDF page 1, formulate the actual Koopman operator on L^2 and orthogonal eigenfunction bases. The directly read primary sources are New's notes and Solomyak's notes.

The scope is explicit: this disproves the necessary infinite-value jump for the printed counting *function*, with Hilbert dimension and real or modulus thresholds. It does not claim to refute a different Hamel-dimension interpretation, an unspecified ordering on arbitrary complex spectra, or a rule sampled only at distinct actual spectral points. The identity example has only one such point. These interpretations are not silently conflated.

The package pins Lean 4.33.0 and Mathlib commit

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